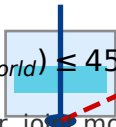




Constraint: $\theta = \cos^{-1}(\hat{z}_{cup} \cdot \hat{z}_{world}) \leq 45^\circ$

Objective: minimize end-effector error, joint motion, and joint-limit proximity



— cup axis
- - 45 deg limit